Pars 2.0 — GPU-Native Sovereign L1 with DEX+EVM+FHE Trifecta

From private-messaging substrate (1.0) to financially-complete sovereign L1 for the Persian diaspora

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Pars Network Launch Retrospective — February 14, 2026

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Abstract

On 14 February 2026 the Pars Network activated its 2.0 epoch: a sovereign, GPU-native Layer 1 carrying the canonical *trifecta* of every real L1 in the Lux family — a native decentralized exchange (DEX), an Ethereum Virtual Machine execution layer (EVM), and a fully-homomorphic encryption substrate (FHE). Pars 1.0 (2025) shipped the messaging, identity, and federated-learning foundation that the diaspora needed but omitted the financial rails; Pars 2.0 closes that gap without rewriting 1.0 primitives. The 2.0 chain runs Quasar consensus 4.0 with QuasarSTM 4.0 execution, enforces the GPU-Residency invariant (LP-137), and participates in the cross-chain Liquidity Protocol launching 20 April 2026. This paper documents the 1.0→2.0 transition, the trifecta integration, the activation receipts captured at block height $h_0 = 14{,}000{,}000$ on the Pars main DAG, and the production parameters now in force.

Keywords: Pars Network, sovereign L1, GPU-native consensus, DEX+EVM+FHE trifecta, Persian diaspora, Liquidity Protocol, post-quantum.

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1 Launch Summary

Field	Value
Network name	Pars Network 2.0
Activation block	$h_0 = 14{,}000{,}000$ on Pars main DAG
Activation timestamp	2026-02-14T00:00:00Z (UTC, mainnet)
Chain ID (C-Chain)	7070 (mainnet), 7071 (testnet)
Consensus	Quasar 4.0 (cf. LP-105)
Execution	QuasarSTM 4.0 (cf. LP-010 v4)
GPU residency	LP-137 enforced
EVM compatibility	Cancun + Pars precompiles 0x2000–0x2300, 0x0700
FHE substrate	CKKS / TFHE hybrid (cf. `pars-fhe-privacy`)
DEX activation	2026-02-14 (genesis pools)
Liquidity Protocol	Onboarding wave 2026-04-20
Native token	PARS, fixed supply 10^9 (no change from 1.0)
Reserve token	ASHA (cf. `pars-asha-reserve`)
Staking lock token	VeASHA (cf. `pars-veasha-tokenomics`)

Table 1: Pars 2.0 launch parameters in force at h_0.

The 2.0 launch is *strictly additive*. No 1.0 primitive was removed. The Session protocol (`pars-session-protocol`), mesh transport (`pars-mesh-networking`), Persian-NLP precompile (`pars-persian-nlp`), federated-learning protocol (`pars-federated-learning`), and post-quantum suite (`pars-post-quantum`) all remain operational with identical on-wire formats. What changed is the addition of three first-class financial subsystems on a GPU-native consensus engine.

2 Pars 1.0 → 2.0 Evolution

2.1 What 1.0 Shipped (2025)

Pars 1.0 was a *dual-layer* architecture (`pars-dual-layer`):

- An EVM-compatible rollup posting state roots to Lux for security inheritance, capable of 2,400 TPS with 1.2-s confirmation.
- A Session daemon layer providing end-to-end encrypted messaging with forward secrecy and deniability, mesh-routable over LoRa/Bluetooth/IP.
- A federated-learning fabric for Persian-language model training without raw-data egress (`pars-federated-lea`
- A PQ migration plan committing to ML-DSA-87, ML-KEM-1024, and SPHINCS+ for keys minted after activation (`pars-post-quantum`).

The diaspora-sovereignty thesis was complete on the *communication* axis. The financial axis was deferred: 1.0 lacked a native DEX, lacked private settlement, and lacked enough throughput in the EVM layer to serve as a primary venue for tokenized assets.

2.2 What 2.0 Adds (2026-02-14)

Three first-class subsystems and one consensus upgrade:

1. **Sovereign L1.** Pars graduates from rollup to standalone L1 running Quasar 4.0 consensus. State commitments to Lux remain available as an *optional* bridging mechanism but are no longer the security root.

2. **Native DEX.** A GPU-native AMM (PARSWAP) with explicit support for the Liquidity Protocol cross-chain settlement API, designed for Persian-rial–denominated and IRT-pegged stablecoin flows.

3. **EVM 4.0.** Cancun-compatible EVM extended with Pars-specific precompiles for calendar conversion (Jalali ↔ Gregorian), Pars Name Service (PNS), and Persian-NLP inference.

4. **FHE substrate.** CKKS / TFHE hybrid scheme exposed as a precompile + sidecar, allowing confidential transactions, private voting, and encrypted federated-learning aggregation (`pars-fhe-privacy`).

2.3 Migration Path for 1.0 Users

The 2.0 genesis state was constructed by the deterministic procedure of Algorithm 1.

Algorithm 1 Pars 1.0 → 2.0 state migration, applied at h_0

```
procedure MigrateState(S_1.0, R_1.0)
    S_2.0 ← ∅
    for each account a ∈ S_1.0 do
        S_2.0[a] ← S_1.0[a]                          ▷ balance, nonce, code, storage
    end for
    S_2.0.ChainID ← 7070                             ▷ Pars C-Chain mainnet
    S_2.0.ConsensusEpoch ← 4
    S_2.0.ExecutionEpoch ← 4
    S_2.0.GPUResidency ← enforced
    S_2.0.Precompiles[LXBook ... LXFeed, FHE] ← active
    S_2.0.FHESubstrate ← CKKS ⊕ TFHE
    S_2.0.LiquidityProtocol ← registered             ▷ onboarding 2026-04-20
    return S_2.0
end procedure
```

Property 2.1 (Account preservation)**.** *For every $a \in S_{1.0}$, the post-migration account $S_{2.0}[a]$ has identical balance, nonce, deployed code, and storage trie root as $S_{1.0}[a]$.*

Property 2.2 (Identity preservation)**.** *Every Pars DID issued under `pars-did-standard` v1.0 verifies unchanged against the 2.0 PDS resolver.*

3 GPU-Native Quasar 4.0 + QuasarSTM 4.0

3.1 Why GPU-Native

Pars 1.0 used CPU-bound execution. The throughput ceiling was set by single-thread EVM interpretation and ECDSA verification. Pars 2.0 inherits the GPU-Residency invariant from Lux 4.0 (LP-137):

Invariant 3.1 (GPU residency)**.** *For every transaction τ executed at height h, the validator's canonical execution path runs on a GPU device g with all input state, intermediate state, and output state resident on g's HBM for the duration of τ. Any movement of state between host RAM and device HBM during τ violates the invariant.*

The invariant is enforced by QuasarSTM 4.0 transactional memory boundaries (cf. LP-010 v4) and is observable via the `gpu_residency_violations` Prometheus counter exported by every Pars validator.

3.2 Throughput at h_0

Metric	Pars 1.0 (2025)	Pars 2.0 (2026-02-14)	Ratio
Sustained TPS (EVM)	2,400	48,000	20×
Confirmation latency	1.2 s	180 ms	6.7×
ECDSA verifications/s	11,000	620,000	56×
ML-DSA-87 verifications/s	—	84,000	—
Block size cap	30M gas	250M gas	8.3×

Table 2: Throughput metrics observed on Pars mainnet validators at $h_0 + 1{,}000$.

These numbers are within 4% of the figures projected in `pars-gpu-scaling` (2025).

4 The Trifecta on Pars

The trifecta is the canonical configuration of a real Lux-family L1: a DEX for value, an EVM for logic, and an FHE substrate for confidential computation, all GPU-native and post-quantum. Pars 2.0 is the third network to adopt the trifecta, after Lux 4.0 (Feb 14) and Hanzo 4.0 (Feb 14).

4.1 DEX — PARSWAP and Liquidity Protocol Integration

PARSWAP is a constant-function automated market maker exposed as a system contract at `0x0000...PA75`. Its design is intentionally narrow: it is the venue for diaspora remittance flows and tokenized real-asset settlement. Three classes of pool are supported:

1. **Stable pools** (StableSwap-style invariant, $A = 200$) for IRT-pegged stablecoins, USDC, USDT, USDP-pegged synthetic rial.
2. **Weighted pools** (Balancer-style $\prod_i x_i^{w_i} = k$) for diversified tokenized-asset baskets.
3. **Concentrated-liquidity pools** (Uniswap v3-style ticks) for high-volume PARS pairs.

Liquidity Protocol bridge. Per the Liquidity Protocol common API (LP-310), every pool exposes a deterministic settlement endpoint that the cross-chain router can call to atomically rebalance liquidity across Pars, Hanzo, Zoo, and Lux DEXes. The bridge wave activates 2026-04-20; the on-chain registration was committed at h_0.

The pool-state update equation for a stable pool is the StableSwap invariant:

$$A \cdot n^n \cdot \sum_{i=1}^{n} x_i + D = A \cdot D \cdot n^n + \frac{D^{n+1}}{n^n \prod_{i=1}^{n} x_i}. \tag{1}$$

Equation 1 is solved on-GPU per swap; the Newton iteration converges in ≤ 6 rounds for all production parameters.

4.2 EVM — Pars Precompiles

The Pars EVM is Cancun-equivalent and reuses the canonical Lux precompile ranges (cf. Lux registry v0.4.x): `0x0600`–`0x06FF` for post-quantum, `0x0700`–`0x07FF` for FHE/privacy, `0x2000`–`0x23FF` for native DEX/HFT, `0x1000`–`0x12FF` for X/T/Z-chain bridges. Pars adds three precompiles at the empty `0x0c00` slot for diaspora-specific functions:

Address	Function	Gas
0x2000	LXBook (HFT orderbook — canonical Lux DEX)	50,000
0x2100	LXPool (AMM pools — canonical Lux DEX)	50,000
0x2300	LXFeed (price feeds — canonical Lux DEX)	5,000
0x0700	FHE (CKKS / TFHE — canonical Lux privacy)	500,000
0x0c01	CalendarConvert (Jalali ↔ Gregorian)	250
0x0c02	PNS_Resolve (Pars Name Service)	1,500
0x0c03	PersianNLP_Tokenize	5,000 + 50/token

Table 3: Canonical and Pars-specific precompiles active from h_0.

The PNS precompile makes domain resolution a constant-time on-chain operation. The Persian-NLP precompile wraps the inference work documented in `pars-persian-nlp` and `pars-distributed-in` PARSWAP is a thin wrapper over LXPool plus a Liquidity-Protocol routing adapter; it is not a fork.

4.3 FHE — Confidential Transactions and Private Voting

Pars 2.0 ships the FHE substrate documented in `pars-fhe-privacy` as a first-class L1 capability rather than a sidecar. Three application patterns are activated at h_0:

1. **Confidential transfers.** Amount-hiding transfers using CKKS-encrypted amounts and a homomorphic balance-update circuit. Sender and recipient are public; amounts are confidential.

2. **Private voting.** Per `pip-0104-private-voting-protocol`, ballots are TFHE-encrypted; tallying is performed under encryption; only the final tally is decrypted by a threshold committee.

3. **Encrypted federated-learning aggregation.** Replaces the 1.0 secure-aggregation step of `pars-federated-learning` with homomorphic gradient summation, removing the need for an honest aggregator.

The FHE precompiles operate on ciphertexts encoded under the threshold public key $\mathsf{tpk}_{\mathrm{Pars}}$ produced by a t-of-n distributed key generation among Pars validators. Decryption requires the cooperation of any t validators; no single validator holds the secret key in clear form.

5 Tokenomics — Preserved and Extended

The 1.0 tokenomics defined in `pars-tokenomics-staking`, `pars-asha-reserve`, and `pars-veasha-tokenomics` are preserved unchanged. Three extensions accompany 2.0:

- **DEX fee routing.** 0.30% of swap volume routes to a fee-distribution contract; the split is 50% liquidity providers, 30% Asha reserve, 15% VeAsha lockers, 5% DAO treasury. This mirrors the canonical Lux-family routing.

- **FHE relay rewards.** A new emission stream, capped at 0.5% of annual PARS supply, rewards validators that perform FHE bootstrapping and ciphertext maintenance. Caps are enforced at the protocol level.

- **Liquidity Protocol incentive vault.** A 2-year vault of 2×10^7 PARS bootstraps cross-chain liquidity in the April 20 wave. Drawdown is gated by snapshot proofs of cross-chain volume.

Theorem 5.1 (Supply preservation)**.** *The 2.0 tokenomics changes preserve the fixed total supply of* 10^9 *PARS established in 1.0. The FHE relay rewards and Liquidity Protocol incentive vault are funded from the previously-allocated ecosystem-incentives bucket, not from new emission.*

Proof. By construction. The 1.0 tokenomics paper allocates 30% of total supply (3×10^8) to ecosystem incentives. The FHE relay reward cap is 0.5% of annual supply for ≤ 8 years $= 0.5\% \times 8 \times 10^9 = 4 \times 10^7$. The Liquidity Protocol incentive vault is 2×10^7. Both fit within 3×10^8. □

6 Cross-Chain Bridges

Pars 2.0 maintains canonical bridges to the three other trifecta networks. Bridge security relies on threshold-signed light-client proofs plus a fallback fraud-proof window of 7 days.

Counterparty	Mechanism	Proof	Settlement
Lux	A-Chain attestation (cf. `lux-achain-attestation`)	ML-DSA-87	24 s
Hanzo	Liquidity Protocol router	ML-DSA-87	30 s
Zoo	Liquidity Protocol router	ML-DSA-87	30 s
Ethereum	Optimistic light client	ECDSA + ML-DSA-87	7 d

Table 4: Pars 2.0 cross-chain bridges. All paths post-quantum at the signature layer.

The Liquidity Protocol router unifies Pars–Hanzo and Pars–Zoo flows under a single message format; the Lux bridge retains its native attestation path because Lux is the root of trust for the family.

7 Diaspora Tooling under Sanctions-Resilient Mesh

Pars 2.0 keeps the diaspora as the primary user. Three new application patterns are unlocked by the trifecta.

7.1 Confidential Remittance

A diaspora user in Toronto sends value to a recipient in Tehran. The flow:

1. Sender mints a CKKS-encrypted transfer to the recipient's stealth address (`pars-stealth-addresses`).
2. Settlement occurs on Pars 2.0 with hidden amount.
3. Recipient claims via threshold decryption involving a Pars-Iran relay node and at least two diaspora-resident validators.
4. Routing over the Pars mesh (`pars-mesh-networking`) provides delivery even when the recipient is under selective IP blocking.

7.2 Tokenized Real-Asset DEX

Real-asset issuance (gold receipts, commercial paper, fractional real-estate units) on the Pars EVM is paired with PARSWAP pools. Liquidity Protocol's cross-chain router exposes the same assets to the Lux, Hanzo, and Zoo communities, multiplying liquidity without replicating issuance.

7.3 Private DAO Voting at Scale

Pars DAO governance (`pars-dao-governance`) moves from clear-text ballots to TFHE-encrypted voting. Ballots are submitted, tallied homomorphically, and the result revealed; individual votes never leave the encrypted domain. Coercion-resistance properties from `pars-coercion-resistance` are preserved and strengthened.

8 Activation Receipts

The launch was block-aligned at $h_0 = 14{,}000{,}000$. The receipts below were captured by the Pars block explorer between h_0 and $h_0 + 5{,}000$.

Event	Height	Notes
Quasar 4.0 epoch boundary	h_0	cert lane id 0x04
QuasarSTM 4.0 activation	h_0	STM root rotated
GPU-Residency invariant on	$h_0 + 1$	first violation counter = 0
PARSWAP genesis pools	$h_0 + 12$	7 stable, 4 weighted, 9 concentrated
FHE threshold key DKG	$h_0 + 24$	$t = 67$, $n = 100$
PNS root commitment	$h_0 + 30$	Merkle root 0x4f...
Liquidity Protocol register	$h_0 + 48$	router id lp.pars.router
First confidential transfer	$h_0 + 91$	tx 0x21...
First private vote tally	$h_0 + 720$	ballots = 4,312

Table 5: Selected mainnet receipts from Pars 2.0 activation.

9 Conclusion

Pars 2.0 promotes the network from a private-messaging substrate to a financially-complete sovereign L1 without abandoning the diaspora sovereignty thesis. The DEX provides the venue, the EVM provides the logic, the FHE substrate provides the confidentiality, the GPU-native consensus provides the speed, and the Liquidity Protocol exposes Pars liquidity to the rest of the Lux family. Every 1.0 primitive remains operational and unchanged.

The April 20, 2026 Liquidity Protocol activation will close the final gap: cross-chain remittance and tokenized-asset flows under a single common API, with Pars as a first-class participant.

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